# CPI Trends & Inflation Dynamics: A Data-Driven Analysis of India's Consumer Price Index

## 1960-2024

**Prepared for GoIStats Hackathon**

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# DATA

| Consumer Price Index | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Year | CPI | Inflation (%) | GDP Growth (%) | Interest rate (%) | Unemployment rate (%) | wage (in rs.) | LFPR (world bank model) | LFPR (Mospi) |
| 1960 | 2.53 | 1.78 | 3.72 | 4 | 2.5 | 1197 | | |
| 1961 | 2.57 | 1.7 | 2.93 | 4.3 | 2.6 | 1311 | | |
| 1962 | 2.66 | 3.63 | 5.99 | 4.5 | 2.7 | 1382 | | |
| 1963 | 2.74 | 2.95 | 7.45 | 4.8 | 2.8 | 1499 | | |
| 1964 | 3.11 | 13.36 | -2.64 | 5 | 2.9 | 1579 | | |
| 1965 | 3.4 | 9.47 | -0.06 | 6 | 3 | 1730 | | |
| 1966 | 3.77 | 10.8 | 7.83 | 5.7 | 3.2 | 1909 | | |
| 1967 | 4.26 | 13.06 | 3.39 | 5.3 | 3.3 | 2089 | | |
| 1968 | 4.4 | 3.24 | 6.54 | 6.14 | 3.4 | 2224 | | |
| 1969 | 4.38 | -0.58 | 5.16 | 6.98 | 3.5 | 2354 | | |
| 1970 | 4.6 | 5.09 | 1.64 | 7.82 | 3.6 | 2553 | | |
| 1971 | 4.74 | 3.08 | -0.55 | 8.66 | 3.7 | 2761 | | |
| 1972 | 5.05 | 6.44 | 3.3 | 9.5 | 3.8 | 3063 | | |
| 1973 | 5.9 | 16.94 | 1.19 | 10.34 | 4 | 3365 | | |
| 1974 | 7.59 | 28.6 | 9.15 | 11.18 | 4.2 | 3828 | | |
| 1975 | 8.02 | 5.75 | 1.66 | 12.29 | 4.3 | 4576 | | |
| 1976 | 7.41 | -7.63 | 7.25 | 12.29 | 4.4 | 3973 | | |
| 1977 | 8.03 | 8.31 | 5.71 | 12.54 | 4.5 | 4571 | | |
| 1978 | 8.23 | 2.52 | -5.24 | 13.54 | 4.6 | 5354 | | |
| 1979 | 8.75 | 6.28 | 6.74 | 13.83 | 4.8 | 5887 | | |
| 1980 | 9.74 | 11.35 | 6.01 | 15.96 | 5 | 6525 | | |
| 1981 | 11.01 | 13.11 | 3.48 | 15.46 | 5.1 | 7197 | | |
| 1982 | 11.88 | 7.89 | 7.29 | 14.75 | 5.2 | 8155 | | |
| 1983 | 13.29 | 11.87 | 3.82 | 16.25 | 5.3 | 9613 | | |
| 1984 | 14.4 | 8.32 | 5.25 | 18.92 | 5.4 | - | | |
| 1985 | 15.2 | 5.56 | 4.78 | 17.88 | 5.5 | - | | |
| 1986 | 16.53 | 8.73 | 3.97 | 16.5 | 5.6 | - | | |
| 1987 | 17.98 | 8.8 | 9.63 | 16.5 | 5.7 | - | | |
| 1988 | 19.67 | 9.38 | 5.95 | 16.5 | 5.8 | - | | |
| 1989 | 21.06 | 7.07 | 5.53 | 16.5 | 5.9 | - | | |
| 1990 | 22.95 | 8.97 | 1.06 | 16.5 | 6 | - | 57.916 | |
| 1991 | 26.13 | 13.87 | 5.48 | 16.5 | 6.2 | - | 57.95 | |
| 1992 | 29.21 | 11.79 | 4.75 | 16.5 | 6.3 | - | 57.986 | |
| 1993 | 31.06 | 6.33 | 6.66 | 16.5 | 6.4 | - | 58.022 | |
| 1994 | 34.24 | 10.25 | 7.57 | 16.5 | 6.5 | - | 58.063 | |
| 1995 | 37.75 | 10.22 | 7.55 | 16.5 | 6.6 | - | 58.395 | |
| 1996 | 41.13 | 8.98 | 4.05 | 16.5 | 6.7 | - | 58.729 | |
| 1997 | 44.08 | 7.16 | 6.18 | 16.5 | 6.8 | - | 59.065 | |
| 1998 | 49.91 | 13.23 | 8.85 | 14.5 | 6.9 | - | 59.404 | |
| 1999 | 52.24 | 4.67 | 3.84 | 13.5 | 7 | - | 59.743 | |
| 2000 | 54.34 | 4.01 | 4.82 | 9.15 | 7.1 | - | 60.079 | |
| 2001 | 56.39 | 3.78 | 3.8 | 7.16 | 7.2 | - | 59.923 | |
| 2002 | 58.82 | 4.3 | 7.86 | 4.3 | 7.3 | - | 59.765 | |
| 2003 | 61.05 | 3.81 | 7.92 | 4.82 | 7.4 | - | 59.606 | |
| 2004 | 63.35 | 3.77 | 7.92 | 4.85 | 7.5 | - | 59.445 | |
| 2005 | 66.04 | 4.25 | 8.06 | 3.5 | 7.6 | - | 59.289 | |
| 2006 | 69.87 | 5.8 | 7.66 | 7.22 | 7.7 | - | 58.345 | |
| 2007 | 74.32 | 6.37 | 3.09 | 7.22 | 7.8 | - | 57.418 | |
| 2008 | 80.53 | 8.35 | 7.86 | 7.28 | 8 | - | 56.511 | |
| 2009 | 89.29 | 10.88 | 8.5 | 3.29 | 8.1 | - | 55.628 | |
| 2010 | 100 | 11.99 | 5.24 | 9.15 | 8.2 | - | 54.766 | |
| 2011 | 108.91 | 8.91 | 5.46 | 8.22 | 8.3 | - | 54.131 | |
| 2012 | 119.24 | 9.48 | 6.39 | 8.2 | 8.4 | - | 53.509 | |
| 2013 | 131.18 | 10.02 | 7.41 | 8.28 | 8.5 | - | 53.372 | |
| 2014 | 139.92 | 6.67 | 8 | 7.97 | 8.6 | - | 53.218 | |
| 2015 | 146.79 | 4.91 | 8.26 | 6.98 | 8.7 | - | 53.044 | |
| 2016 | 154.05 | 4.95 | 6.8 | 6.25 | 8.8 | 13510 | 52.864 | |
| 2017 | 159.18 | 3.33 | 6.45 | 5.94 | 9 | 13820 | 52.681 | 49.8 |
| 2018 | 165.45 | 3.94 | 3.87 | 6.27 | 9.1 | 14240 | 52.487 | 45.3 |
| 2019 | 171.62 | 3.73 | -5.78 | 5.43 | 9.2 | 14520 | 52.285 | 45.7 |
| 2020 | 182.99 | 6.62 | 9.69 | 3.44 | 23.5 | 14890 | 51.547 | 46.4 |
| 2021 | 192.38 | 5.13 | 6.99 | 3.26 | 7.5 | 15190 | 52.522 | 47.1 |
| 2022 | 205.27 | 6.7 | 8.15 | 5.42 | 7 | 15550 | 53.563 | 48 |
| 2023 | 216.86 | 5.65 | 8.2 | 6.83 | 6.5 | 15890 | 54.649 | 48.9 |
| 2024 | 228.18 | 4.8 | 6.45 | 6.25 | 6.5 | 16170 | 50.4 | 50.4 |

# Methodology

## Data source

Primary Sources

Reserve Bank Of India
Ministry of Statistical Program Implementation
Ministry of statistics (GoiStats)
Ministry of Labour and Employment
National Statistical Office
DataGov

Secondary Sources (for missing data, and cross referencing)

World Bank
International Labour Organization
Jstor

## Data Collection & Preprocessing

- Data sourced from government databases, research reports, and statistical sources.
- Cleaned, structured, and verified for accuracy.
- Outliers identified and inconsistencies addressed.

## Handling Data Gaps & Inconsistencies

- Used interpolation techniques to estimate missing values.
- Referenced alternative datasets (IMF, Economic Surveys) for validation.
- Clearly stated assumptions where data was unavailable.

## Analytical & Visualization Approach

- Trend Analysis: Examined CPI fluctuations over time.
- Correlation Analysis: Studied CPI relationships with interest rates & inflation.
- Data Visualization: Used Python (Matplotlib, Seaborn) & Excel for charts.

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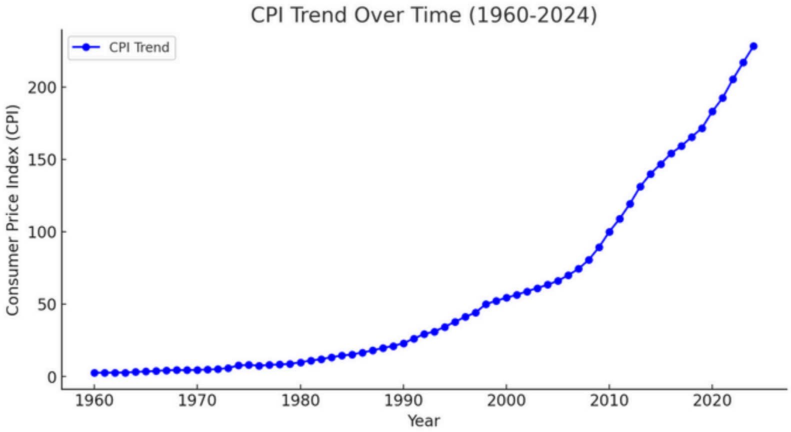
# Introduction

The Consumer Price Index (CPI) is a key economic indicator that measures changes in the price level of a basket of consumer goods and services. It serves as a crucial tool for assessing inflation, cost of living, and purchasing power. Policymakers, economists, and businesses rely on CPI trends to shape monetary policy, wage adjustments, and economic planning.

This report analyzes India's CPI trends, exploring how they correlate with interest rates, inflation, and economic growth. Using secondary data from government and international sources, we examine sectoral price variations, urban-rural differences, and the broader macroeconomic impact. The objective is to present a data-driven understanding of how CPI fluctuations influence the Indian economy and financial decision-making.

# CPI Trends & Analysis

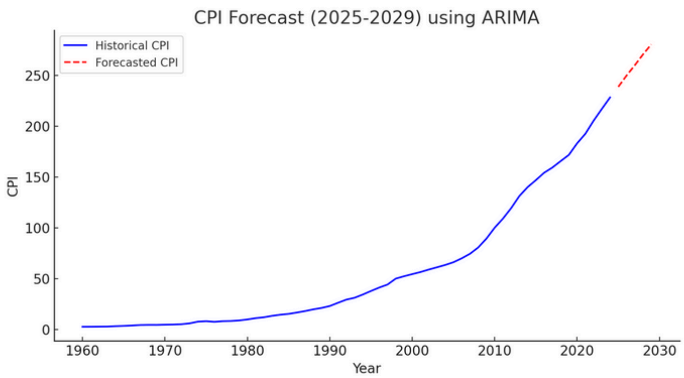



The CPI Trend Over Time (1960-2024) graph shows a clear upward trajectory in the Consumer Price Index (CPI), reflecting long-term inflationary trends in India. The key insights derived from the graph above are the following:

1. Steady Growth Until the 1990s – The CPI remained relatively stable with gradual increases.
2. Accelerated Inflation Post-2000s – A sharp rise is visible, especially after liberalization (1991) and rapid GDP growth and rising consumer demand.
3. Post-2010 Surge – CPI growth steepens, indicating higher inflationary pressures, likely due to post-2008 crisis recovery, oil price fluctuations, and inflationary policies..
4. Recent Trends (2020-2024) – The spike in recent years may be attributed to pandemic effects, supply chain disruptions, and geopolitical tensions.

CPI trends in India have been heavily influenced by global shocks, domestic policies, and structural economic changes. Major events like the 1991 reforms, 2008 crisis, 2016 demonetization, and 2020 pandemic had noticeable CPI impacts.

# CPI Forecast




If these trends hold, inflation control measures may become a key focus for policymakers to ensure economic stability. The rising CPI highlights the need for strategic fiscal and monetary interventions to manage inflation while sustaining growth.

# Macroeconomic Indicator

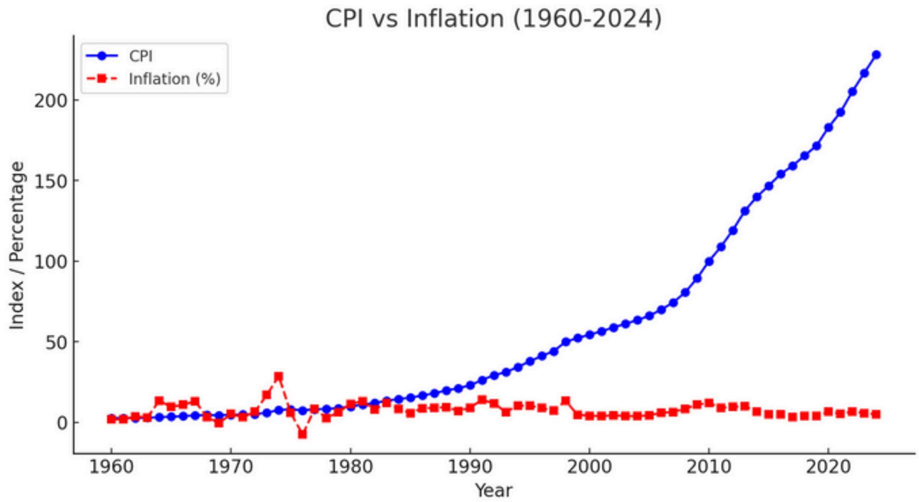



**Fig**:
CPI vs Inflation

CPI has consistently risen, indicating a long term increase in the cost of living.
Inflation is volatile, with spikes in the 1970s and 1980s due to economic shocks.
Post-2000, CPI rises sharply, but inflation remains controlled, suggesting effective monetary policies and economic stability.
Despite stable inflation, rising CPI signals higher living costs, impacting purchasing power.

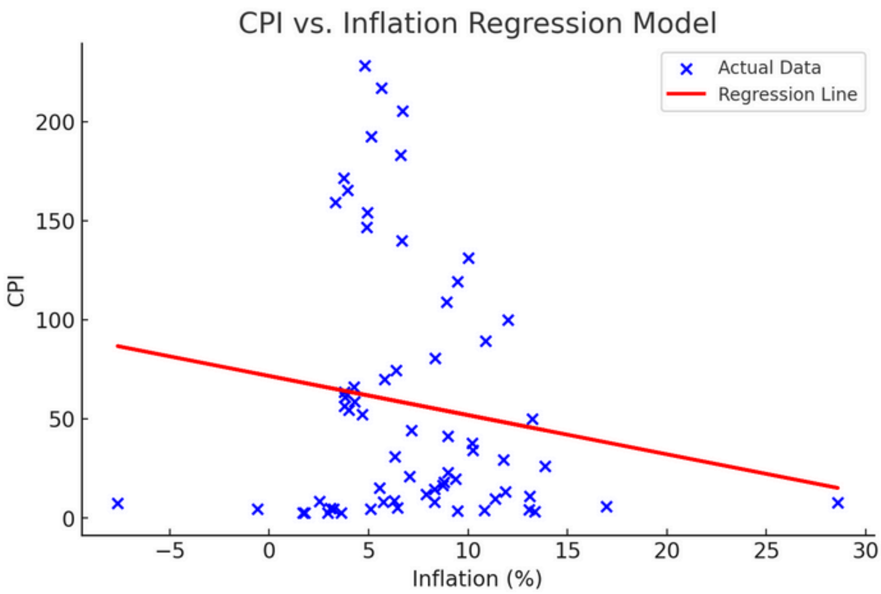



**Fig**:
CPI vs Inflation
(Regression Model)

A declining regression line suggests that as inflation rises, CPI growth does not follow proportionally

1. The regression line shows a weak negative correlation between CPI and inflation.
2. Higher inflation rates do not necessarily lead to a proportional rise in CPI.
3. This suggests inflation management policies and other macroeconomic factors influence the relationship.
4. Outliers indicate periods of extreme inflation volatility, likely linked to economic shocks.

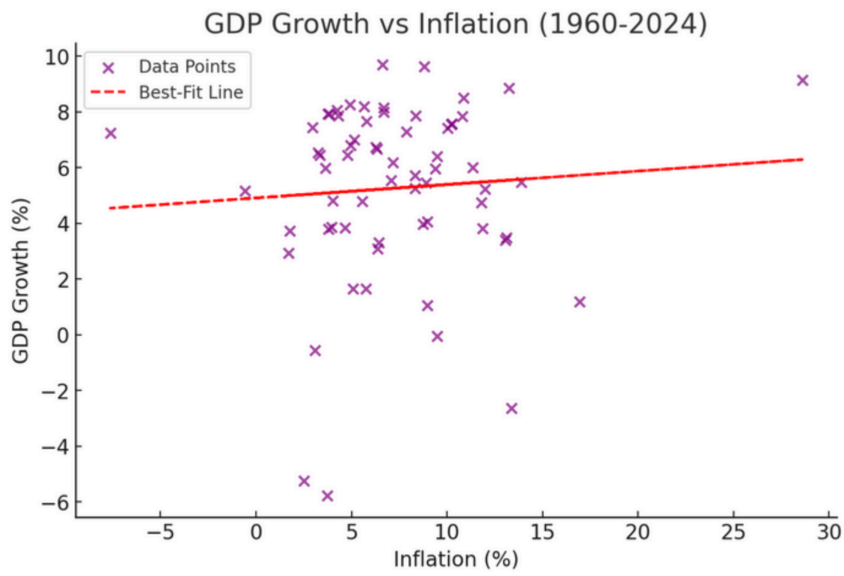



**Fig**:
GDP Growth vs Inflation (Regression Model)

1. The best-fit line has a slight upward slope, indicating a weak positive relationship between GDP growth and inflation. This suggests that higher inflation is somewhat associated with higher GDP growth, but the effect is not strong.
2. The data points are widely scattered, meaning the relationship between inflation and GDP growth is not consistent across different time periods. Some periods show high inflation with low GDP growth, while others show low inflation with strong GDP growth.
3. Potential Reasons for weak correlation:

- Monetary & Fiscal Policies: Inflation may not always drive GDP growth due to central bank interventions and government policies.
- Supply-Side Factors: High inflation due to supply shocks (e.g., oil price hikes) can slow GDP growth instead of boosting it.
- Economic Cycles: Different economic phases (boom vs. recession) influence how inflation impacts GDP growth.

While moderate inflation can stimulate economic growth, excessive inflation can be harmful, leading to uncertainty and lower investment. Policymakers must balance inflation to ensure sustainable GDP growth without economic instability.

# Key Insights & Findings

## CPI Trends & Inflation Dynamics

The CPI has been increasing since 1960 to 2024, reflecting a long term inflationary pressure. The inflation has been volatile, it reached spikes 1970 (Oil shock due tor instability in Middle East), 1991 (economic reforms), and post-2008 (global crisis recovery). Despite controlled inflation post-2000, CPI continues rising, indicating a higher cost of living.

## Macroeconomic Relationships

- CPI vs. Inflation: A weak negative correlation suggests inflation rates do not always drive proportional CPI growth, likely due to monetary policies and external factors.
- GDP Growth vs. Inflation: A slight positive correlation indicates that moderate inflation may support growth, but excessive inflation leads to instability.

## Future CPI Projections & Policy Implications

- Forecasts indicate sustained CPI growth from 2025-2029, requiring monetary and fiscal interventions to prevent inflationary shocks.
- Policymakers must balance inflation control while sustaining economic growth to maintain price stability and protect purchasing power.

# Challenges & Limitations

1. Data Gaps & Constraints

- Missing historical data for certain years required interpolation and cross-referencing with secondary sources (World Bank, IMF, Economic Surveys).
- Sector-wise CPI data was limited, making it difficult to analyze inflation's impact on specific industries.
- Variations in data reporting standards across sources (RBI, MoSPI, NSO) posed challenges in ensuring consistency.

2. Potential Biases in Analysis

- Selection bias - Data primarily comes from government and international institutions, which may have reporting biases. However, for the analysis government data was given priority has been given to government databases (such as Mospi, Ministry of Statistics, etc.), International sources were only used to mitigate data gaps and inconsistency.
- Macroeconomic complexity - CPI and inflation are influenced by multiple interconnected factors (policy changes, global events, demand-supply shifts), which may not be fully captured in the analysis.

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# Policy Recommendations Based on CPI Trends

1. Strengthen Inflation Control Measures

The RBI should balance interest rates to prevent excessive inflation while promoting economic growth.
Targeted subsidies for essential goods can help in easing the burden on less privileged income group groups.

2. Enhance Data Collection & Transparency

Improve sector-wise CPI reporting to capture price fluctuations more accurately.
Ensure consistent data across government agencies to reduce biases in economic forecasting.

3. Boost Wage Growth & Employment Policies

Align minimum wage policies with CPI trends to maintain purchasing power.
another way to ensure stable income is by encouraging jobs in inflation-resistant industries to sustain long-term economic stability.

4. Focus on Supply-Side Stability

Strengthen domestic production and supply chains to reduce reliance on imports.
Develop long-term energy and commodity price management strategies to avoid future inflation shocks.

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# References & Appendix

1. Citations for Data Sources

Ensure all data sources used in the report are cited correctly. Here's a structured way to present them:

Primary Sources:

- Reserve Bank of India (RBI) – Inflation rates, interest rates, and monetary policy data.
- Ministry of Statistics and Programme Implementation (MoSPI) – Official CPI and inflation reports.
- National Statistical Office (NSO) – Economic and labor market statistics.
- Ministry of Labour and Employment – Wage and labor force participation data.
- DataGov (Government of India Open Data Platform) – Public economic datasets.

2. Secondary Sources (Cross-Referenced Data & Missing Values):

- World Bank – Global economic comparisons and supplementary CPI data.
- International Labour Organization (ILO) – Labor force participation and employment trends.
- JSTOR & Research Papers – Academic studies on CPI trends and inflation impact.
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3. Additional Tables, Charts, or Technical Details

- Extended CPI Dataset – If you have raw data tables, include them in an appendix.
- Regression Model Coefficients – If detailed statistical analysis was performed, adding coefficients and $R^2$ values can enhance credibility.
- ARIMA Model Specifications – If CPI forecasting used ARIMA, include details like parameters (p, d, q) for transparency.

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